

Roll No.

TCS-331

B. TECH. (CSE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

FUNDAMENTAL OF IoT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any **one** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Smart Parking, powered by NB-IoT technology, is making it easier for drivers to find free parking spots. Cities can better manage their parking assets and maximise the revenue available to them as a result. Drivers searching for parking create congestion and pollution by circling and hunting for available parking. Smart Parking services are able to significantly ease these problems by guiding a driver directly to a parking space. Design a IoT based framework to ensure that only authorised vehicles park in these spaces. (CO1)

P. T. O.

OR

- (b) A smart city is a municipality that uses information and communication technologies (ICT) to increase operational efficiency, share information with the public and improve both the quality of government services and citizen welfare. Analyse the applications of IoT in Smart Cities.

(CO1)

2. (a) Total Crane Solutions adopted CheckedOK to keep track of inspections but now uses it to manage its entire operation. Automating tasks has enabled its founder to build a growing business while keeping overheads to a minimum. Explore the Components of an RF.D system in perspective of Total Crane Solutions.

(CO1)

OR

- (b) At Cork University Hospital, an early warning system has been put into action to monitor staff constantly for elevated temperature. This warning system uses a sensor that identifies staff that is starting to develop symptoms. They can then be notified to stay home and rest. Recognize the communication protocols of Internet of Things have to used to design an IoT based system at Cork University Hospital. (CO1)
3. (a) Ayla Networks provides the industry's first Agile IoT Platform, accelerating development, support, and ongoing enhancements of connected products for the Internet of Things. Ayla's software fabric runs across devices, cloud, and apps to create secure connectivity, data analytics, and feature-rich customer experiences. Apply the Key IoT Technologies and design a framework for Ayla Networks. (CO2)

OR

(b) Ofo bikes was founded in China in 2014 and currently operates over 10 million dockless hire bikes in over 180 cities around the world. In China, Ofo operates in all major cities including Beijing, Shanghai and Guangzhou. They have partners with China Telecom in China to improve their operations and make their hire bikes easier to use and lower the cost of managing them. Is it possible to horizontally scale IoT devices of Ofo ? Compare and contrast with vertical scaling of IoT devices. (CO2)

4. (a) Zara is a retail clothing brand that uses RFID technology to manage inventory in the warehouse and to trace clothes in the store. It aids to deal with overstocking and under stocking digitally and also updates the availability of particular clothes at the stores. Examine the different types of RFID Security challenges in the perspective of Zara. (CO2)

OR

(b) Bosch provides the products and services for Drive and Control Technology, Packaging Technology, Power tools for Professionals, Security Solutions, and Software Solutions. Bosch provides the IoT Suite for connecting and managing devices, sensors, and gateways. It provides secure access management. It also provides an open IoT platform for all domains. Classify the IoT sensors based on input, output and datatype in the perspective of Bosch. (CO2)

P. T. O.

5. (a) The education sector or education system is a group of institutions (ministries of education, local educational authorities, teacher training institutions, schools, universities, etc.) whose primary purpose is to provide education to children and young people in educational settings. Describe the role of WSN in education sector along with its architecture. (CO2)

OR

- (b) Mozilla WebThings is an open source implementation of emerging Web of Things standards at the W3C. W3C Web of Things is an initiative that aims to reduce the IoT fragmentation, through the recently launched Web of Things Working Group. W3C started to develop the initial standards for the Web of Things, aiming to reduce the costs of development, lessen the risks to both investors and customers, and encourage exponential growth in the market for IoT devices and services. Relate the Web things with the Internet of Things. (CO2)